Global Change Biology (2009) **15**, 976–991, doi: 10.1111/j.1365-2486.2008.01744.x

Quantifying photosynthetic capacity and its relationship to leaf nitrogen content for global-scale terrestrial biosphere models

JENS KATTGE*, WOLFGANG KNORR†, THOMAS RADDATZ‡ and CHRISTIAN WIRTH*
*Max-Planck-Institute for Biogeochemistry, Hans-Knöll Street 10, 07745 Jena, Germany, †QUEST, Department of Earth Sciences, University of Bristol, Wills Memorial Building, Queen's Road, BS8 1RJ, UK, ‡Max Planck Institute for Meteorology, Bundesstraße 53, 20146 Hamburg, Germany

Abstract

Photosynthetic capacity and its relationship to leaf nitrogen content are two of the most sensitive parameters of terrestrial biosphere models (TBM) whose representation in global-scale simulations has been severely hampered by a lack of systematic analyses using a sufficiently broad database. Here, we use data of qualitative traits, climate and soil to subdivide the terrestrial vegetation into functional types (PFT), and then assimilate observations of carboxylation capacity, V_{max} (723 data points), and maximum photosynthesis rates, A_{max} (776 data points), into the C_3 photosynthesis model proposed by Farquhar *et al.* to constrain the relationship of V_{max}^{25} (V_{max} normalised to 25 °C) to leaf nitrogen content per unit leaf area for each PFT. In a second step, the resulting functions are used to predict V_{max}^{25} per PFT from easily measurable values of leaf nitrogen content in natural vegetation (1966 data points). Mean values of V_{max}^{25} thus obtained are implemented into a TBM (BETHY within the coupled climate–vegetation model ECHAM5/JSBACH) and modelled gross primary production (GPP) is compared with independent observations on stand scale. Apart from providing parameter ranges per PFT constrained from much more comprehensive data, the results of this analysis enable several major improvements on previous parameterisations. (1) The range of mean V_{max}^{25} between PFTs is dominated by differences of photosynthetic nitrogen use efficiency (NUE, defined as V_{max}^{25} divided by leaf nitrogen content), while within each PFT, the scatter of V_{max}^{25} values is dominated by the high variability of leaf nitrogen content. (2) We find a systematic depression of NUE on certain tropical soils that are known to be deficient in phosphorous. (3) V_{max}^{25} of tropical trees derived by this study is substantially lower than earlier estimates currently used in TBMs, with an obvious effect on modelled GPP and surface temperature. (4) The root-mean-squared difference between modelled and observed GPP is substantially reduced.

Keywords: A_{max}, carbon–nitrogen interaction, data assimilation, Farquhar model, global carbon cycle, leaf nitrogen content, Markov Chain Monte Carlo sampling, phosphorous limitation, photosynthesis, V_{max}

Received 23 April 2008; revised version received 24 July 2008 and accepted 4 September 2008

Introduction

Global terrestrial biosphere models (TBM) commonly include the photosynthesis scheme developed by Farquhar *et al.* (1980). This model has proven to be particularly useful in the context of environmental change, because it calculates photosynthesis based on a mechanistic representation depending, for example, on atmospheric CO_2 concentration, leaf temperature and nitrogen content. However, results are particularly sensitive to the parameterisation of photosynthetic capacity, which substantially contributes to uncertainties of model predictions (White *et al.*, 2000; Knorr & Heimann, 2001; Knorr & Kattge, 2005; Zaehle *et al.*, 2005). Furthermore, reliable estimates of the relationship of photosynthetic capacity to leaf nitrogen content are needed in models of coupled carbon and nitrogen budgets of the terrestrial biosphere, which are indispensable for a better understanding and quantification

Correspondence: Jens Kattge, tel. +49 3641 57 6226, fax +49 3641 57 7200, e-mail: jkattge@bgc-jena.mpg.de

© 2009 The Authors
Journal compilation © 2009 Blackwell Publishing Ltd

of the CO_2 fertilisation effect, which is assumed to be limited by nitrogen availability (Hungate *et al*., 2003; Reich *et al*., 2006; Thornton *et al*., 2007).

Progress in parameterising TBMs has been hampered by the fact that photosynthetic capacity is still the parameter that is most difficult to prescribe on a global scale, as it can only be measured at the leaf level (Reich *et al*., 2007) where it exhibits a range of almost two orders of magnitude (Wullschleger, 1993; Medlyn *et al*., 1999, this study). The difficulty of finding enough data has previously been avoided by relying on isotopic leaf tissue data (Beerling & Quick, 1995). Here, a different approach will be used, based on qualitative traits, climate, soil and the correlation of photosynthetic capacity to leaf nitrogen content (Medlyn *et al*., 1999), which facilitates a significant improvement in the number of measurements that can be exploited for the parameter estimation, as well as the treatment of uncertainties.

There is now a firm empirical basis for this kind of trait-based approach. Reich *et al*. (2007) have recently shown that maximum rates of photosynthesis, A_{max}, based on leaf dry mass can be predicted from simple qualitative plant traits like phylogeny, plant growth form and leaf phenology, in combination with climate information and easily measurable quantitative traits. Within this study, we use basic qualitative traits (phylogeny, plant growth form, leaf phenology and morphology) combined with information of climate and soil to stratify the terrestrial vegetation into 10 plant functional types (PFTs). Apart from the specification along soil types, this approach has been employed in numerous TBMs, for example, the Sheffield (Woodward *et al*., 1995) and LPJ (Sitch *et al*., 2003) global dynamic vegetation models, and in coupled climate–vegetation models such as HadCM3LC based on TRIFFID (Cox *et al*., 2004) or ECHAM5/JSBACH (Raddatz *et al*., 2007) based on BETHY (Knorr, 2000). We, here, additionally included the separation along soil type, as old tropical soils are known to be deficient in phosphorous with a potential impact on the relationship of photosynthetic capacity to leaf nitrogen content (Meir *et al*., 2007).

On a leaf area basis, which is the more general reference for use in TBMs and therefore the one of interest in our analysis, leaf nitrogen content is the quantitative trait which offers most information to predict A_{max} (Wright *et al*., 2004). Accordingly, the relationship between photosynthetic capacity and leaf nitrogen content is comparatively stable under different environmental conditions: height within the canopy (Ellsworth & Reich, 1993; Meir *et al*., 2002), season of the year (Misson *et al*., 2006), concentration of atmospheric CO_2 during plant growth (Medlyn *et al*., 1999; Ellsworth *et al*., 2004) and for different species of the same PFT (Carswell *et al*., 2000; Kattge, 2002; Ellsworth *et al*., 2004).

In the photosynthesis model proposed by Farquhar *et al*. (1980), photosynthetic capacity is defined by the parameters 'carboxylation capacity at standard temperature' (V_{max}^{25}) and 'electron transport capacity at standard temperature' (J_{max}^{25}). While both parameters are tightly correlated, carboxylation capacity, V_{max}, has been measured much more frequently (Wullschleger, 1993; Leuning, 1997; Kattge & Knorr, 2007). However, V_{max} in living tissue cannot be measured directly but must be inferred by model inversion from photosynthesis measurements. Therefore, instead of exclusively using V_{max} from published model inversions, it is possible to perform standardised model inversions on available measurements of the maximum photosynthetic rate, A_{max}, which is usually derived under favourable natural conditions (light saturation, favourable temperatures, no obvious restriction due to soil water content or vapour pressure deficit). Until now, both data sources, V_{max} and A_{max}, have only been used for mutual cross-validation (Peterson *et al*., 1999; Ellsworth *et al*., 2004), but they have not yet been systematically combined.

The two novel aspects of the approach sketched out so far are the use of qualitative traits and climate in combination with soil information to stratify the terrestrial vegetation into PFTs, and the combined use of V_{max} and A_{max} data to constrain the relationship of V_{max}^{25} to leaf nitrogen content per PFT. The first enables the analysis to account for phosphorous limitation, which is likely to occur on old tropical soils, and the second facilitates a high number of measurements to constrain the relationship of V_{max}^{25} to leaf nitrogen content per PFT. The constrained relationship is then used to predict V_{max}^{25} per PFT globally from a larger dataset of leaf nitrogen content in natural vegetation. Errors of parameters defining the relationships of V_{max}^{25} to leaf nitrogen content per PFT are propagated and consistently combined with errors of data for leaf nitrogen content in natural vegetation. This error propagation is important if results shall be used as prior information within assimilation schemes at the site (Knorr & Kattge, 2005) and global scales (Rayner *et al*., 2005). Finally V_{max}^{25} values are implemented into a TBM and modeled gross primary production (GPP) is compared with stand-scale observations.

Material and methods

Overview

The parameter estimation framework includes the following steps (Fig. 1): (1) All V_{max} data are standardised to one consistent formulation of the photosynthesis model by Farquhar *et al*. (1980) and vegetation is subdivided into PFTs. (2) The functional relationship of V_{max}^{25} to leaf nitrogen content is optimised for each

© 2009 The Authors

13652486, 2009, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2486.2008.01744.x by Luonnonvarakeskus, Wiley Online Library on [17/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

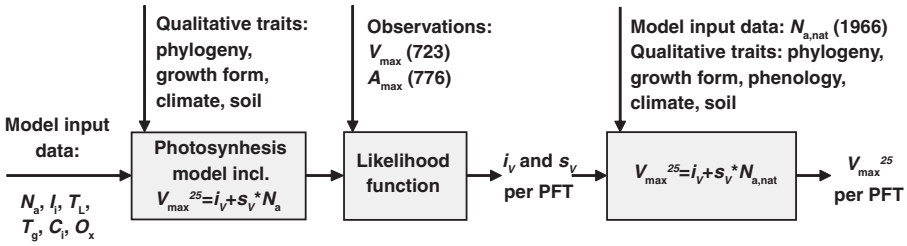

Fig. 1 Assimilation scheme to optimise parameters (i_V, s_V) for the relationship of V_{max}^{25} to leaf nitrogen content (N_a) against observations of V_{max} and A_{max} and then predict V_{max}^{25} per PFT from observations of leaf nitrogen content in natural vegetation ($N_{a,nat}$). N_a, leaf nitrogen content; I_i, incident radiation; T_l, leaf temperature during measurement; T_g, mean plant growth temperature; C_i and O_x, intercellular CO_2 and O_2 concentrations; $N_{a,nat}$, leaf nitrogen content of natural vegetation. Number of V_{max} and A_{max} data points used for the optimisation of i_V and s_V and number of $N_{a,nat}$ data points used for the derivation of V_{max}^{25} are shown in parentheses.

PFT based on data of V_{max} and A_{max} from leaves with known nitrogen content measured in natural or experimental environment (*assimilation step*). (3) The constrained relationship of V_{max}^{25} to leaf nitrogen content is used to infer V_{max}^{25} per PFT from a much larger dataset of leaf nitrogen content of natural vegetation, denoted $N_{a,nat}$ (*prediction step*). Here, we also derive mean and uncertainties of nitrogen use efficiency of carboxylation per PFT (NUE, calculated as $V_{max}^{25}/N_{a,nat}$).

Predicted means of V_{max}^{25} are compared with a previously published global dataset (Beerling & Quick, 1995). Both sets of V_{max}^{25} are used in the TBM BETHY within the coupled climate–vegetation model ECHAM5/JSBACH (Knorr, 2000; Raddatz *et al.*, 2007). The impact of model parameterisation on modelled GPP is evaluated at the scale of the model grid and compared with observations at the stand scale (Luyssaert *et al.*, 2007).

Photosynthesis model

Global TBMs usually employ a rather traditional formulation of the photosynthesis model, neglecting leaf CO_2 transfer conductance (Ethier *et al.*, 2006) and the decline of photosynthetic capacity with leaf age (Escudero & Mediavilla, 2003). Here, we chose a formulation of the Farquhar *et al.* (1980) model following Medlyn *et al.* (2002), with a parameterisation of Rubisco kinetics and temperature dependence of photosynthesis according to Bernacchi *et al.* (2001), which has recently been extended to account for an acclimation of photosynthesis to plant growth temperature for the application in global models (Kattge & Knorr, 2007).

Carboxylation capacity at the reference temperature of 25 °C, V_{max}^{25}, is assumed to be linearly related to leaf nitrogen content, N_a, (Medlyn *et al.*, 1999)

$$V_{max}^{25} = i_V + s_V \times N_a \tag{1}$$

with $V_{max}^{25}$and N_a expressed per unit leaf area. The relationship is determined by the intercept, i_V, and the slope, s_V, which are assumed to be specific for each PFT.

To include measurements of A_{max}, we calculate net leaf CO_2 uptake, A_n, as

$$A_n = \min(J_C, J_E) - R_d, \tag{2}$$

where J_C is the rate of photosynthesis when Rubisco activity is limiting, J_E the rate when ribulose-1,5-bisphosphate (RuBP)-regeneration is limiting and R_d the mitochondrial respiration in daylight. R_d at 25 °C is assumed being a constant fraction of V_{max}^{25}(Farquhar *et al.*, 1980). At environmental conditions typical for measurements of A_{max} (light saturation, favourable temperatures, ambient CO_2 concentrations) photosynthesis can be assumed as limited by J_C, described as

$$J_C = V_{max}\frac{C_i - \Gamma_*}{C_i + K_c(1 + O_x/K_o)}, \tag{3}$$

where V_{max} is the maximum carboxylation rate at leaf temperature, K_c and K_o are the Michaelis–Menten coefficients of RuBP carboxylation and oxygenation, Γ_* is the CO_2 compensation point in the absence of mitochondrial respiration, and C_i and O_x are the concentrations of CO_2 and O_2, respectively, within the leaf air spaces. V_{max} is related to the model parameter V_{max}^{25} (V_{max} at standard temperature of 25 °C), based on a commonly used expression for the temperature dependency of V_{max}^{25} (Medlyn *et al.*, 2002), extended to account for an empirically derived acclimation to plant growth temperatures of preceding days to weeks, T_g, (Kattge & Knorr, 2007):

$$V_{max} = V_{max}^{25}\exp\left[\frac{E_a(T_l - T_{ref})}{(T_{ref}RT_l)}\right] \frac{1 + \exp\{[T_{ref}(a_{\Delta S} - b_{\Delta S} \times T_g) - H_d]/(T_{ref}R)\}}{1 + \exp\{[T_l(a_{\Delta S} - b_{\Delta S} \times T_g) - H_d]/(T_lR)\}}. \tag{4}$$

Standardisation of V_{max} *and stratification of vegetation into PFTs*

Data of V_{max} derived from gas exchange measurements depend on the parameterisation of Rubisco kinetics

© 2009 The Authors

13652486, 2009, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2486.2008.01744.x by Luonnonvarakskus, Wiley Online Library on [17/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

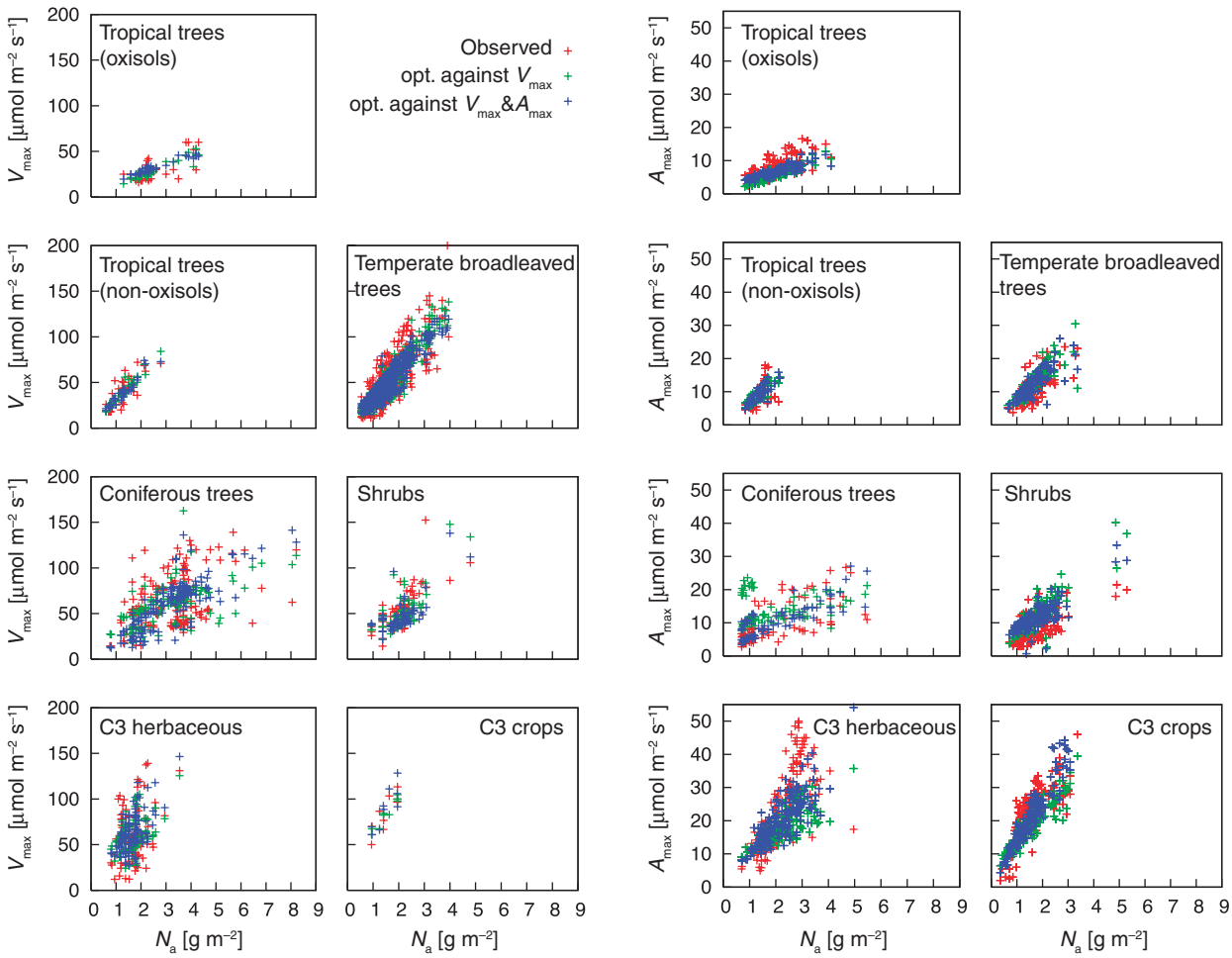

Fig. 2 Model-data comparison of V_{max} (left hand side), and A_{max} (right hand side) as function of N_a for different plant functional types. Red: observed values; green: modelled values optimised against V_{max} data only; blue: modelled values optimised against V_{max} and A_{max} data.

used within the Farquhar *et al*. (1980) model (Medlyn *et al*., 2002). Therefore, all V_{max} data have been standardised to one consistent formulation and parameterisation of the Farquhar *et al*. (1980) model following the method described by Kattge & Knorr (2007).

Stratification of vegetation into PFTs was based on the qualitative traits phylogeny (angiosperm, coniferous gymnosperm), plant growth form (tree, shrub, herb, grass or crop) and leaf phenology (evergreen, deciduous) in combination with information about climate (tropical, temperate, boreal) and soil type (measured individuals predominantly grown on oxisols, which are old tropical soils with especially low phosphorous availability, or grown on non-oxisol soils). Leaf phenology may be taken as a qualitative proxy for the quantitative trait leaf lifespan and the division along soil type enables the analysis to account for phosphorous limitation.

Within the assimilation step, we could only reasonably distinguish between seven PFTs (given in Table 2 and Fig. 2) due to a limited number of data, not accounting for leaf phenology. However, more data were available for inferring PFT specific ranges of V_{max}^{25} from leaf nitrogen content of natural vegetation and we could distinguish between 10 PFTs (given in Table 3 and Fig. 3), also accounting for leaf phenology. Therefore, temperate broadleaved trees, coniferous trees and shrubs each are assigned one set of parameters for i_V and s_V, together with their probability distributions, irrespective of whether they are evergreen or deciduous; when inferring V_{max}^{25} from observed $N_{a,nat}$ we then differentiate between evergreen and deciduous PFTs.

Data assimilation step

The data assimilation step consists of deriving probability distributions for the parameters i_V and s_V. It is conducted as follows (see Fig. 1). We selected gas exchange studies of V_{max} and A_{max} that also report leaf nitrogen content per leaf area, N_a. Using the leaf temperature during the measurement, T_l, the plant growth

© 2009 The Authors

13652486, 2009, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2486.2008.01744.x by Luonnonvarakskus, Wiley Online Library on [17/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

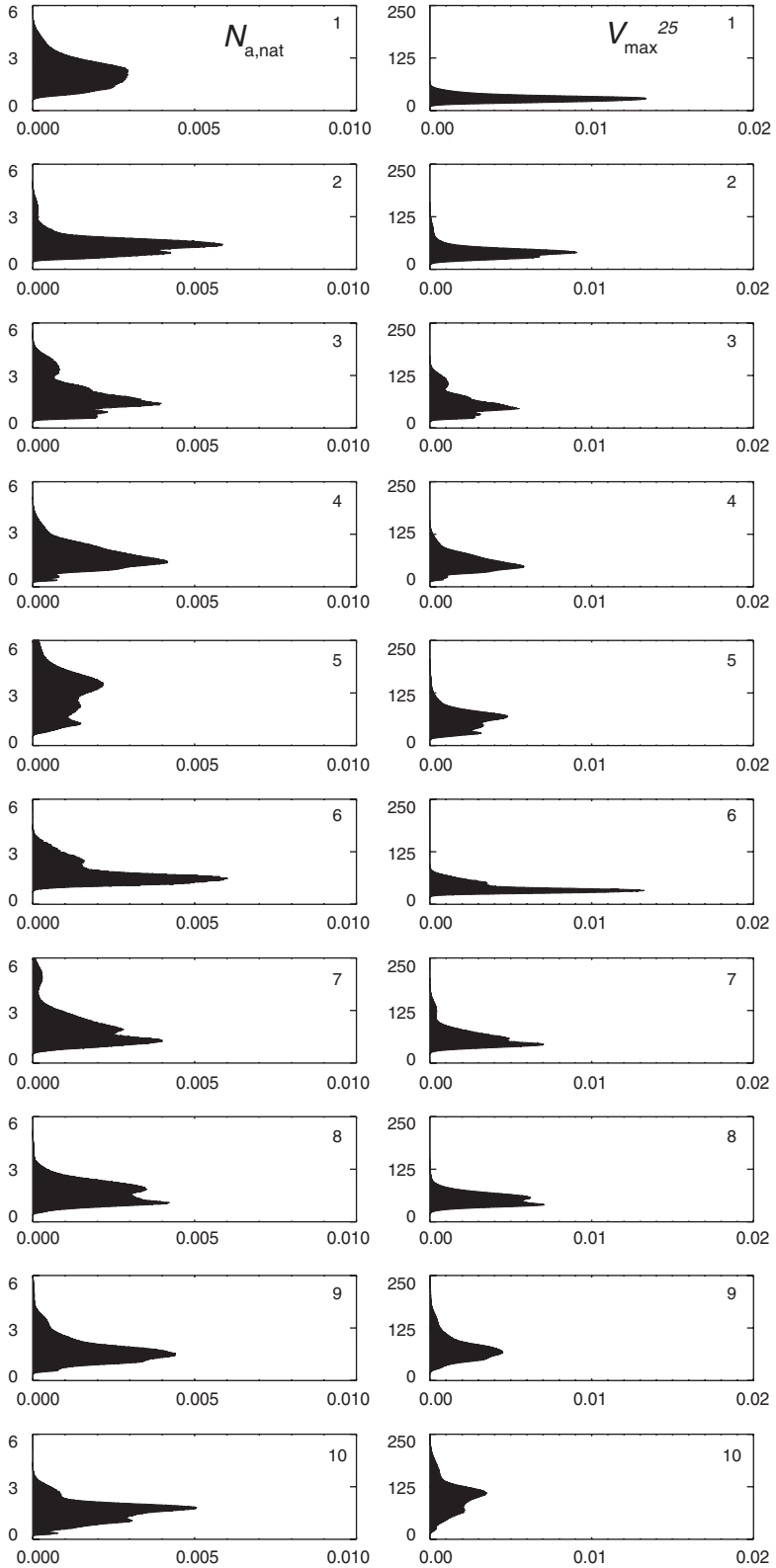

Fig. 3 Histograms of probability density distributions of observed leaf nitrogen content of natural vegetation ($N_{a,nat}$, g m^{-2}, left panels) and predicted photosynthetic capacity (V_{max}^{25}, μmol m^{-2} s^{-1}, right panels) for different PFTs. (1) Tropical trees predominantly on oxisols, (2) tropical trees on non oxisols, (3) temperate broadleaved evergreen trees, (4) temperate broadleaved deciduous trees, (5) evergreen coniferous trees, (6) deciduous coniferous trees, (7) evergreen shrubs, (8) deciduous shrubs, (9) C3 herbaceous, (10) C3 crops. Histograms set to 1000 intervals.

© 2009 The Authors

13652486, 2009, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2486.2008.01744.x by Luonnonvarakeskus, Wiley Online Library on [17/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

temperature, T_g, the intercellular concentration of CO_2 and O_2, C_i and O_x, respectively, and the incident irradiance, I_i, derived from the publications or external sources, we relate flux-measurements of A_{max} to the model parameter V^{25}_{max}. In a Bayesian inversion approach using a Markov Chain Monte Carlo (MCMC) sampling technique, we minimise the mismatch between modelled and observed V_{max} and A_{max}, and at the same time sample the probability distribution of the optimised parameters i_V and s_V (the so-called posterior probability density function) for each PFT. Details can be found in Knorr & Kattge (2005). We optimised the parameters once against V_{max} data only (V_{max} approach) and against V_{max} and A_{max} (V_{max} and A_{max} approach) to determine if the two approaches yield consistent results.

We account for uncertainties of input data by randomly choosing one realisation drawn from their Gaussian probability density function at each proposed step of the Markov Chain. Errors due to missing or incorrect process description and parameterisation of the Farquhar *et al*. (1980) model itself are not considered in this study. A typical MCMC simulation generates about 150 000 samples for input data and parameters.

Inferring PFT-specific V^{25}_{max} (Prediction step)

In the following step, the sampled probability distribution of parameters i_V and s_V from the MCMC simulation is used to infer V^{25}_{max} and its uncertainty ranges for each PFT from a large number of samples of leaf nitrogen content of natural vegetation, $N_{a,nat}$ (see Fig. 1). This dataset is much larger than the datasets used for the MCMC simulations, even though only natural vegetation is included. We account for the uncertainty of $N_{a,nat}$ data by randomly sampling from a Gaussian distribution. The resulting sample is then grouped into PFTs, and for each PFT we derive the mean and standard deviation. The NUE of carboxylation, calculated as $V^{25}_{max}/N_{a,nat}$, is derived simultaneously.

Model input data and observations

The following data are used as model input to calculate values of V_{max} and A_{max}: nitrogen content per leaf area (N_a and $N_{a,nat}$), leaf temperature during measurement (T_l), plant growth temperature (T_g), intercellular CO_2 concentration (C_i), intercellular O_2 concentration (O_x). Incident irradiance (I_i) is necessary to verify that photosynthesis is limited by carboxylation (Eqn (2)). Data of V_{max} and A_{max} are used as observations to be assimilated, as both are also predicted either by Eqns (1) and (4), or by the complete photosynthesis model. A list of references to observations and leaf nitrogen content of natural vegetation is given in Table 1.

Table 1 References to measurements of carboxylation capacity (V_{max}) and maximum photosynthesis rate (A_{max}) used to constrain the relationship of V^{25}_{max} to leaf nitrogen content and references of measurements leaf nitrogen content in natural environment ($N_{a,nat}$) used to predict V^{25}_{max} per PFT

Reference	PFT	V_{max}	A_{max}	$N_{a,nat}$
Bassow & Bazzaz (1997)	4		x	x
Carswell *et al*. (2000)	1	x		x
Coste *et al*. (2005)	1		x	x
Domingues *et al*. (2005)	1		x	
Ellsworth *et al*. (2004)	4, 5, 6, 7, 8, 9	x		x
Ellsworth & Reich (1993)	4		x	x
Evans (1989)	10, 4		x	x
Field & Mooney (1986)	9, 8, 7		x	x
Franco *et al*. (2005)	1, 2		x	x
Garnier *et al*. (1999)	9		x	
Grassi *et al*. (2002)	3	x		
Grueters (1998)	10	x		x
Han *et al*. (2004)	5	x		x
Herppich *et al*. (2002)	7		x	x
Hikosaka *et al*. (2002)	1, 7		x	x
Kattge (2002)	3, 4, 5, 8, 9	x		x
Kenzo *et al*. (2004)	2		x	x
Körner (1989)	9		x	x
Kumagai *et al*. (2006)	2	x		x
Le Roux *et al*. (1999)	4	x		x
Lusk & Reich (2000)	4, 5			x
Medlyn & Jarvis (1999)	4, 5	x		x
Medlyn *et al*. (1999)	3, 4, 5, 7	x		x
Medlyn *et al*. (2002)	5	x		x
Meir *et al*. (2002)	2, 4, 5	x		x
Peterson *et al*. (1999)	1, 2, 4, 5, 10		x	x
Quilici & Medina (1998)	2		x	x
Reich *et al*. (1995)	4, 5		x	x
Reich *et al*. (1998)	4, 5, 6		x	
Reich *et al*. (1999)	1, 3, 4, 5, 6, 7, 8, 9,		x	x
Ripullone *et al*. (2003)	4, 5	x		
Sage *et al*. (1987)	9		x	
Schulze *et al*. (1994)	1, 2, 3, 4, 5, 6, 7, 9, 9, 10			x
Springer *et al*. (2005)	4, 5		x	x
Tan & Hogan (1995)	5		x	
Tissue *et al*. (1995)	9	x		
Tissue *et al*. (1999)	5	x		x
Walcroft *et al*. (1997)	4, 5	x		x
Walcroft *et al*. (2002)	4	x		x
White *et al*. (2000)	4, 5, 6, 7, 8, 9			x
Whitehead *et al*. (2004)	8	x		x
Wohlfahrt *et al*. (1999)	7, 9	x		x

1: tropical trees (oxisols); 2: tropical trees (non oxisols); 3: temperate broadleaved evergreen trees; 4: temperate broadleaved deciduous trees; 5: evergreen coniferous trees; 6: deciduous coniferous trees; 7: evergreen shrubs; 8: deciduous shrubs; 9: C3 herbaceous (incl. grass); 10: C3 crops;
PFT: plant functional type.

© 2009 The Authors

13652486, 2009, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2486.2008.01744.x by Luonnonvarakeskus, Wiley Online Library on [17/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

As mentioned earlier, input data are treated as Gaussian distributed. For leaf nitrogen content we assumed an uncertainty of 10% of the mean, in terms of one standard deviation (SD). Leaf temperature during measurements was often reported in the references, while plant growth temperature was rarely given. If temperatures were not reported, they were requested from the authors or estimated from climate tables provided by meteorological stations related to the given region. If temperatures were reported, their uncertainty was assumed to be 0.5 °C. If temperatures were requested or reconstructed, their uncertainty in terms of one SD was assumed to be 2.5 °C. The intercellular CO_2 concentration (C_i) was assumed to be 80% of the atmospheric CO_2 concentration, as a reasonable approximation for conditions during measurements of A_{max}, characterised by full sunlight, favourable temperatures and no water limitation (Yoshie, 1986; Morison, 1987). The uncertainty of C_i was assumed to be 10%. For O_x, a value of 21% was chosen, and no uncertainties were taken into account. Incident irradiance, I_i, necessary to verify carboxylation limitation, was mostly given. If I_i was not available, but measurements were conducted under full sunlight and light saturation was reported, we assumed an incident irradiance of 1800 $\mu mol\, m^{-2} s^{-1}$ (Larcher, 1994).

The uncertainty of V_{max} was assumed to be 10 $\mu mol\, m^{-2} s^{-1}$ (Manter & Kerrigan, 2004), and the one for A_{max} 5 $\mu mol\, m^{-2} s^{-1}$. If the ratio of A_{max} to N_a was below a threshold of 3 $\mu mol\, CO_2\, g\, N^{-1} s^{-1}$ (for temperate trees, herbaceous plants and crops), or below a threshold of 2 $\mu mol\, CO_2\, g\, N^{-1} s^{-1}$ (for tropical trees, coniferous trees and shrubs), we assumed that stomata were partially closed during gas exchange measurements due to water stress and the corresponding data of A_{max} were excluded from the analysis. The photosynthetic capacity, expressed as V_{max} or A_{max}, plotted against leaf nitrogen content shows a heteroskedastic distribution, meaning that the variances of V_{max} and A_{max} increase with N_a (Fig. 2). This is observed in most data compilations of V_{max} and A_{max} against nitrogen content (Field & Mooney, 1986; Medlyn *et al.*, 1999; Peterson *et al.*, 1999). To account for this observed heteroskedasticity, which implies an increasing uncertainty of the single data with increasing leaf nitrogen content, uncertainties of V_{max} and A_{max} were scaled by the value of the respective leaf nitrogen content. Observations had sometimes been derived from several replicated measurements, nevertheless, we did not treat them as multiple observations, but treated all observations equally, because details of variation of repeated measurements were often not described in sufficient detail in the publications.

The number of data points of leaf nitrogen content in natural vegetation ranges from 27 (deciduous coniferous trees) to 404 (temperate broadleaved deciduous trees) per PFT (Table 3). As in the previous step, all data points were treated equally, although each data point often represents several measurements (e.g. data from Schulze *et al.*, 1994). This means that the real number of measured leaves is several times higher than the number of data points. The mean values of $N_{a,nat}$ per PFT varied by a factor of about 2, from 1.41 $g\, m^{-2}$ for tropical trees on nonoxisols to 3.10 $g\, m^{-2}$ for evergreen coniferous trees. The standard deviation of data per PFT was about 30–60% of the respective mean. The standard error per PFT (standard deviation divided by square-root of n) is about 2–10% of the respective mean, depending on the amount of data. The distributions of $N_{a,nat}$ were positively skewed within all PFTs (Table 3). The exact shapes of the distributions are illustrated by histograms in Fig. 3.

Impact of parameterisation on modelled carbon fluxes

Predicted mean values of V_{max}^{25} per PFT are compared with a global parameterisation of V_{max}^{25} derived from $\delta^{13}C$ and A_{max} data (Beerling & Quick, 1995). Both parameter sets of V_{max}^{25} are implemented in the terrestrial biosphere model BETHY (Knorr, 2000) within the coupled climate–vegetation model ECHAM5/JSBACH (Raddatz *et al.*, 2007). A standard set-up similar to the one used in the framework of the IPCC and described in Raddatz *et al.* (2007) is used to simulate a reference period of 20 years with contemporary climate. The ranges of modelled GPP are compared with observed stand-scale data for relevant forest vegetation types within the climatic envelopes representative for the corresponding observed sites given by Luyssaert *et al.* (2007).

Results

Effect of including A_{max} *measurements*

The relationship between V_{max}^{25} and N_a was derived according to Eqn (1), and the results are shown in Table 2. Inclusion of A_{max} measurements in addition to V_{max} data confirms the results derived from V_{max} data alone for most PFTs, in the sense that the uncertainty ranges of the interception and slope parameters, i_V and s_V, overlap between the two optimisations. Also, the ranges of modelled data were similar (Fig. 2). Because of the inclusion of A_{max} data, the uncertainty of i_V (s_V) was reduced by between 11% (13%) for tropical trees on nonoxisols and 68% (65%) for crops (see Table 2). Only in the case of temperate broadleaved trees did the addition of A_{max} data not reduce the uncertainty ranges, due to the large number of data for V_{max} (358) relative to

© 2009 The Authors

13652486, 2009, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2486.2008.01744.x by Luonnonvarakeskus, Wiley Online Library on [17/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 2 Intercept (i_V) and slope (s_V) for V^{25}_{max} as a function of leaf N content per PFT from the optimization against V_{max} data only and against V_{max} and A_{max} data

			V_{max} data only					V_{max} and A_{max} data						
PFT	n_{Vmax}	n_{Amax}	i_V	SD	s_V	SD	corr.	i_V	SD	err.red.	s_V	SD	err.red.	corr.
Tropical trees (oxisols)	33	89	1.99	5.14	10.71	1.90	−0.93	8.90	4.10	0.20	9.30	1.62	0.15	−0.94
Tropical trees (nonoxisols)	33	50	6.35	5.52	25.88	5.16	−0.93	4.19	4.89	0.11	26.19	4.47	0.13	−0.93
Temperate broadleaved trees	358	90	5.40	1.74	30.38	1.34	−0.90	5.73	1.83	−0.05	29.81	1.41	−0.05	−0.91
Coniferous trees	136	77	34.05	5.90	9.71	2.26	−0.91	6.32	2.78	0.53	18.15	1.48	0.34	−0.83
Shrubs	59	188	4.61	8.22	30.20	4.77	−0.93	14.71	4.61	0.44	23.15	3.16	0.34	−0.92
C3 herbaceous	95	158	23.74	6.87	28.17	4.67	−0.96	6.42	4.90	0.29	40.96	3.21	0.31	−0.95
C3 crops	9	124	22.22	16.74	41.27	12.53	−0.96	4.71	5.35	0.68	59.23	4.42	0.65	−0.91

n_{Vmax}: number of datapoints of V_{max}; n_{Amax}: number of datapoints of A_{max}; i_V: intercept ($\mu mol\, CO_2\, m^{-2}\, s^{-1}$); s_V: slope ($\mu mol\, CO_2\, g\, N^{-1}\, s^{-1}$); corr.: correlation coefficient of i_V and s_V; SD: standard deviation; err.red.: relative reduction of uncertainty due to the addition of A_{max} data.

Table 3 Means and statistical moments for observed leaf nitrogen content in natural environment ($N_{a,nat}$, $g\, m^{-2}$), predicted carboxylation capacity at 25 °C (V^{25}_{max}, $\mu mol\, m^{-2}\, s^{-1}$) and nitrogen use efficiency of carboxylation capacity derived by division of V^{25}_{max} by $N_{a,nat}$ (NUE, $\mu mol\, CO_2\, g\, N^{-1}\, s^{-1}$

		$N_{a,nat}$					V^{25}_{max}						NUE			
PFT	$n_{Na,nat}$	Mean	SD	SE	Sk	Ku	Mean	SD	SE	Sk	Ku	BQ	Mean	SE	Sk	Ku
1 Tropical trees (oxisols)	371	2.17	0.80	0.04	0.66	0.30	29.0	7.7	0.4	0.61	0.56	*62	14.02	2.26	1.72	4.72
2 Tropical trees (nonoxisols)	107	1.41	0.56	0.05	1.76	5.41	41.0	15.1	1.5	1.88	6.45	**94	29.60	2.54	0.54	2.45
3 Temperate broadleaved. evergreen trees	65	1.87	0.93	0.11	0.88	0.14	61.4	27.7	3.4	0.89	0.18	41	33.75	2.32	1.4	3.00
4 Temperate broadleaved deciduous trees	404	1.74	0.71	0.04	0.77	0.78	57.7	21.2	1.1	0.78	0.83	35	33.79	2.37	2.94	14.93
5 Evergreen coniferous trees	220	3.10	1.35	0.09	0.74	1.38	62.5	24.7	1.7	0.77	1.53	29	20.72	1.78	1.38	3.93
6 Deciduous coniferous trees	27	1.81	0.64	0.12	1.08	0.49	39.1	11.7	2.3	1.08	0.61	53	22.05	1.61	0.53	0.61
7 Evergreen shrubs	130	2.03	1.05	0.09	1.60	2.65	61.7	24.6	2.2	1.68	3.19	52	32.09	4.24	0.64	1.23
8 Deciduous shrubs	179	1.69	0.62	0.05	0.61	0.47	54.0	14.5	1.1	0.67	0.76	160	33.14	4.38	1.27	3.21
9 C3 herbaceous	254	1.75	0.76	0.05	1.42	2.94	78.2	31.1	2.0	1.44	3.10	42	45.29	2.57	1.79	8.83
10 C3 crops	***209	1.62	0.61	0.04	0.41	0.31	100.7	36.6	2.5	0.43	0.40	120	62.75	3.65	3.13	27.42

$n_{Na,nat}$: number of data, SD and SE: standard deviation and standard error, Sk: skewness, Ku: kurtosis, BQ: estimates of V^{25}_{max} by Beerling & Quick (1995), for comparison. *Tropical evergreen trees; **Tropical deciduous trees; ***All N_a data. PFT, plant functional type.

A_{max} (90). After optimisation against V_{max} and A_{max}, i_V is positive in all cases, but in only three this is significant (Table 2, mean value ± twice standard deviation does not include zero). The slope, s_V, is significantly greater than zero in all cases. Intercept and slope are highly negatively correlated for all PFTs, with a correlation coefficient in the range from −0.83 to −0.95. Note that it is one of the strengths of the MCMC method that the error dependence of both parameters is directly accounted for.

PFT specific V^{25}_{max}

Table 3 gives the mean values and first moments of predicted V^{25}_{max} specified for the 10 PFTs. These, and all other results presented from now on, were derived using both V_{max} and A_{max} data. V^{25}_{max} increases from tropical trees on oxisols (29.0 $\mu mol\, m^{-2}\, s^{-1}$), via tropical trees on nonoxisol soils and deciduous coniferous trees (41.0 and 39.1 $\mu mol\, m^{-2}\, s^{-1}$, respectively) and a group of temperate trees, coniferous trees and shrubs with V^{25}_{max} in the range of 54.0–61.7 $\mu mol\, m^{-2}\, s^{-1}$, to herbaceous C3 plants and crops with very high values of V^{25}_{max} (78.2 and 100.7 $\mu mol\, m^{-2}\, s^{-1}$, respectively). Figure 3 shows histograms of the distributions of predicted V^{25}_{max} per PFT, based on observed data of $N_{a,nat}$ and on the posterior probability density distribution of i_V and s_V. While most PFTs show an irregular distribution of $N_{a,nat}$ data, the histograms of V^{25}_{max} show a tighter probability density distribution than $N_{a,nat}$ for most PFTs.

© 2009 The Authors

13652486, 2009, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2486.2008.01744.x by Luonnonvarakeskus, Wiley Online Library on [17/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

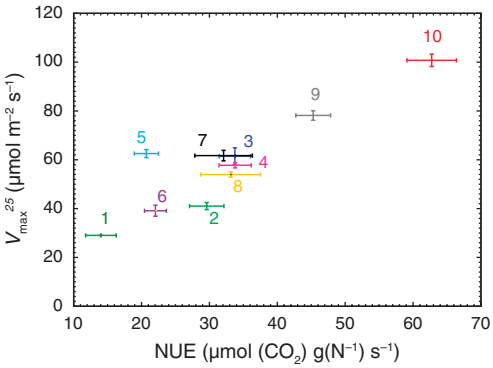

Fig. 4 Plant functional types (PFT) average V_{max}^{25} (µmol m^{-2} s^{-1}) against nitrogen use efficiency (NUE, µmol CO_2 g N^{-1} s^{-1}) defined as $V_{max}^{25}/N_{a,nat}$, with error bars indicating the uncertainty in the global mean (standard error). The numbers refer to the same plant functional types as listed in Table 3 or in the Fig. 3.

Table 3 additionally presents NUE of carboxylation, calculated as $V_{max}^{25}/N_{a,nat}$. The mean values of NUE per PFT indicate three separate groups: low NUE of 14–22 µmol CO_2 g N^{-1} s^{-1} for tropical trees on oxisols, broadleaved and evergreen coniferous trees (PFTs 1, 5, 6); a middle class of NUE of 30–33 µmol CO_2 g N^{-1} s^{-1} by tropical trees on nonoxisols, temperate trees and shrubs (PFTs 2, 3, 4, 7, 8); and high NUE of 45–62 µmol CO_2 g N^{-1} s^{-1} for C3 herbaceous plants and crops (PFT 9, 10; see Table 3). Figure 4 shows that mean values of V_{max}^{25} are linearly related to mean NUE across all PFTs, with the exception of PFT 5 (evergreen coniferous trees: low NUE but high V_{max}^{25}) and PFT 2 (tropical trees on nonoxisol soils: high NUE but low V_{max}^{25}).

Impact of parameterisation in a global TBM

When the V_{max}^{25} values are compared with values derived by Beerling & Quick (1995) for vegetation types and mapped onto PFTs by Knorr & Heimann (2001), major differences emerge for several PFTs, without a consistent bias into any one direction (Table 3). While the means of tropical trees, deciduous coniferous trees and deciduous shrubs are much lower in our approach, temperate trees, evergreen coniferous trees and C3 herbaceous plants show higher means. These differences are obvious in several cases and a notable effect on GPP modelled by the combined climate and TBM ECHAM5/JSBACH-BETHY is observed when the new parameter set of V_{max}^{25} is applied (Fig. 5). There is a pronounced reduction in the productivity gradient going from high to low latitudes. With the new data, tropical forests appear markedly less productive while temperate and boreal forests between 30°N and 60°N have become more productive. While the productivity gradient is remarkably reduced, the global sum of modelled GPP is only slightly changed by the new parameterisation from 181.8 to 163.0 Gt C yr^{-1}.

Comparing modelled GPP with observations for major forest biomes compiled by Luyssaert *et al.* (2007) shows a substantial reduction of mismatch between modelled and observed GPP by the new set of parameters (Table 4, Fig. 6). Using the parameterisation derived by Beerling & Quick (1995) predicts GPP for tropical forests 35% higher than observed and systematically lower for all other biomes (−26% to −62%). In contrast, the new set of V_{max}^{25} predicts lower GPP for tropical forests (−21%), very good approximations for Mediterranean, temperate and boreal humid forests (+3% to +6%), and higher estimates for boreal and temperate arid biomes (+30% and +22%). Notably, the agreement is best where the number of observations in the Luyssaert *et al.* (2007) database is highest, and the largest deviation is found where there are only two datapoints of observed GPP (boreal semiarid deciduous forests, −71%). The sample size of site observations is important, because V_{max}^{25} estimated in this study represents a mean value per PFT, while each site observation is a single case study. The primary productivity of the two boreal semiarid deciduous forests compiled by Luyssaert *et al.* (2007) may indeed be atypically high for this biome (Sebastian Luyssaert, personal communication). Whereas Luyssaert *et al.* (2007) estimated a mean net primary productivity (NPP) of 539 g m^{-2} yr^{-1}, Schulze *et al.* (1999) found a typical value of 123 g m^{-2} yr^{-1} for Siberian deciduous boreal forests, and based on five studies they report a NPP range from 123 to 417 g m^{-2} yr^{-1} for boreal deciduous forests.

Overall, the root mean square error (RMSE) between observed and modelled GPP is almost halved (from 37% to 20%) compared with Beerling & Quick (1995). Excluding the boreal semiarid deciduous forests, the mean RMSE based on the new parameterisation is only about 13% of observed GPP.

Discussion

Global parameterisation of V_{max}^{25}

Commenting on the high variability of photosynthetic NUE, Woodward *et al.* (1995) concluded that photosynthetic capacity of PFTs cannot be determined satisfactorily from leaf-level measurements on a global scale. Since then, a large number of new studies have reported either parameter values of V_{max} or measurements of leaf gas exchange in combination with measurements of leaf nitrogen content. These measurements have been used in the past to validate the Farquhar *et al.* (1980) model (Ellsworth *et al.*, 2004),

© 2009 The Authors

13652486, 2009, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2486.2008.01744.x by Luonnonvarakeskus, Wiley Online Library on [17/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

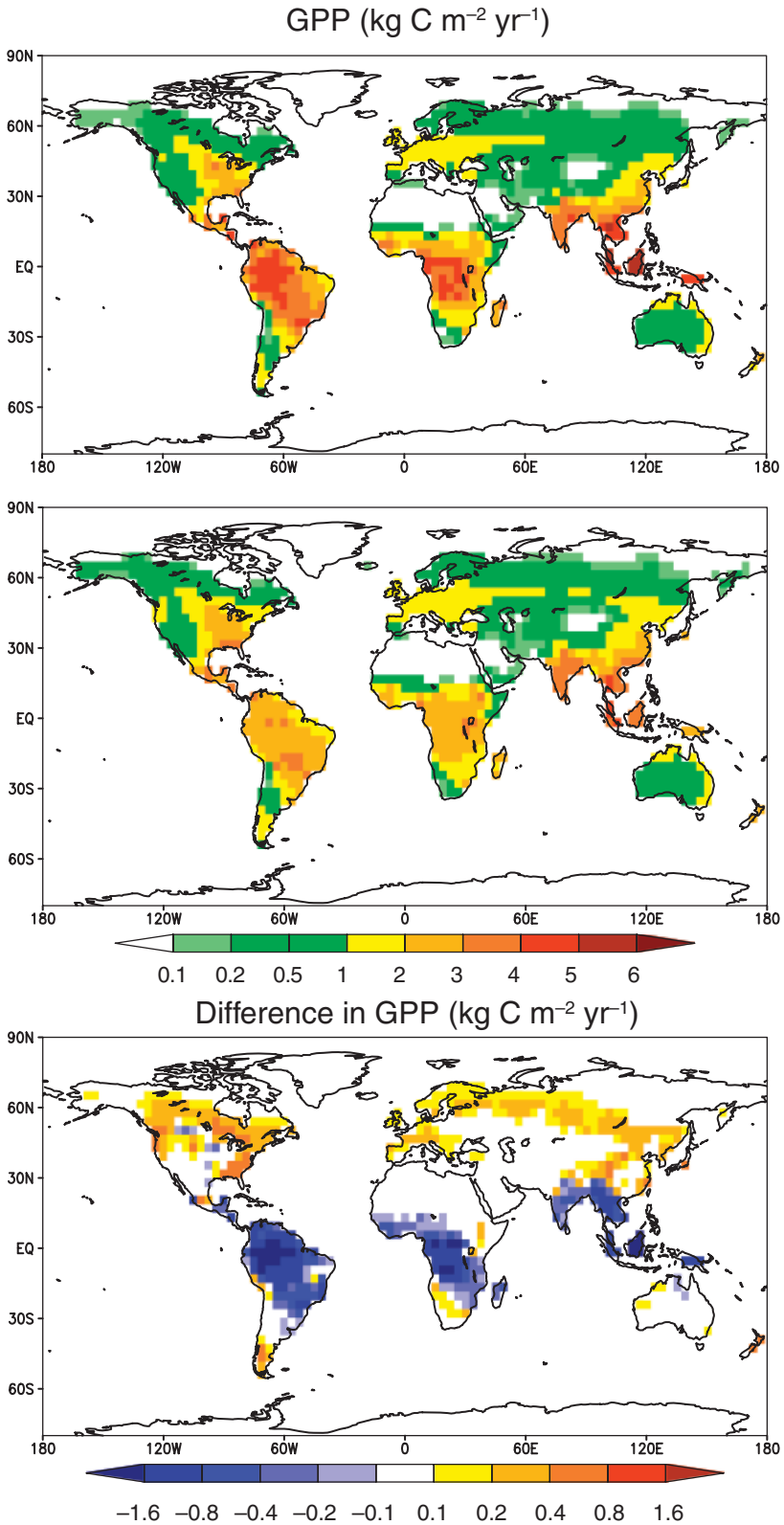

Fig. 5 Twenty-year averages of gross primary production (GPP) (kg C m^{-2} yr^{-1}) in contemporary climate simulated by the coupled climate–vegetation model ECHAM/JSBACH-BETHY after implementation of V_{max}^{25} according to Beerling & Quick (1995) (on top), according to this study (middle) and at the bottom the GPP difference due to the two parameterisations (this study minus Beerling & Quick (1995)).

© 2009 The Authors

13652486, 2009, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2486.2008.01744.x by Luonnonvarakeskus, Wiley Online Library on [17/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 4 Comparison of GPP ($g\,C\,m^{-2}\,yr^{-1}$) for different forest biomes observed on stand scale as compiled by Luyssaert *et al.* (2007) (GPP LY) to GPP modelled on grid scale derived by the coupled climate-vegetation model ECHAM/JSBACH-BETHY using V_{max}^{25} according to this study (GPP) and according to Beerling & Quick (1995) (GPP BQ)

Forest biome	Dominant PFT	*n*	GPP LY	GPP	Error (%)	GPP BQ	Error (%)
Tropical humid evergreen	Tropical trees	6	3551	2814	−21	4790	35
Mediterranean warm evergreen	Temperate broadleaved evergreen trees	8	1478	1526	3	842	−43
Temperate humid evergreen	Temperate broadleaved evergreen trees	29	1762	1834	4	1078	−39
Temperate humid deciduous.	Temperate broadleaved deciduous trees	22	1375	1435	4	1016	−26
Boreal humid evergreen	Evergreen coniferous trees	15	973	1035	6	611	−37
Boreal semi-arid evergreen	Evergreen coniferous trees	7	773	1005	30	572	−26
Temperate semi-arid evergreen	Evergreen coniferous trees	4	1228	1502	22	851	−31
Boreal semi-arid deciduous	Deciduous coniferous trees	2	1201	346	−71	451	−62
	RMSE				20		37

'Forest biome' related to Luyssaert *et al.* (2007), 'dominant PFT' related to selected grid-cells in ECHAM/JSBACH-BETHY, '*n*' number of observations in the database compiled by Luyssaert *et al.* (2007). The deviation of observed and modelled GPP is given in percentage of the observed value (error), summarized over biomes by the root mean squared error (RMSE) over all forest biomes presented.
GPP, gross primary production.

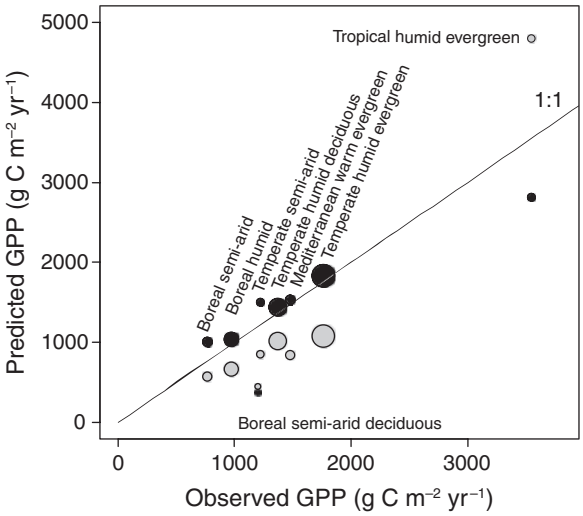

Fig. 6 Validation of gross primary production (GPP) simulated by the coupled climate–vegetation model ECHAM/JSBACH-BETHY with a site-based dataset developed by Luyssaert *et al.* (2007). Grey: V_{max}^{25} according to Beerling & Quick (1995). Black: V_{max}^{25} according to this study. Circle sizes represent the number of sites (2–29, see Table 4) compiled in the Luyssaert *et al.* (2007) dataset.

but have never been exploited jointly to systematically constrain parameter values of photosynthetic capacity. Here, we have been able to show that by combining direct information about parameter values and measurements of A_{max} with easily available information on qualitative plant traits and leaf nitrogen content, these data can be used to constrain NUE and to derive means and uncertainty ranges of photosynthetic capacity for different PFTs. The additional A_{max} data were generally consistent with V_{max} data and substantially reduced the uncertainty of parameters i_V and s_V for most PFTs.

For the outcome of a model parameter estimation exercise like this, reasonable error estimates are an important condition (Raupach *et al.*, 2005; Trudinger *et al.*, 2007). We are confronted here with two major sources of error: measurement errors of observations and error of model input data. Both were taken into account with the methodology used. In the cost functions, we accounted for measurement errors and heteroskedastic distribution of observations. Combining the Markov chain with Monte Carlo sampling, we accounted for uncertainties of model input data. However, we did not explicitly consider error-covariances of observations, which may have resulted in underestimating posterior errors. We accounted for this and for the fact that most assumptions about the errors have to be first order approximations due to lack of explicit information, by using rather large, conservative uncertainty estimates for all data entering the calculation of V_{max} and A_{max} within the assimilation and prediction step.

A first obvious feature of the derived linear relationships of V_{max}^{25} to N_a is the positive intercept, which implies a positive carboxylation capacity for zero leaf nitrogen content. This has also been observed within other studies (Medlyn *et al.*, 1999) and hints at the linear model not being valid at levels of N_a below the observed physiological range. However, there is also good evidence for a saturating relationship, meaning that the fraction of leaf nitrogen allocated to photosynthesis decreases with increasing leaf nitrogen content (Kattge, 2002). Here, we must note that a more complex function that may account for this decrease in

© 2009 The Authors

13652486, 2009, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2486.2008.01744.x by Luonnonvarakeskus, Wiley Online Library on [17/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

allocation cannot be reasonably parameterised by the compiled data, as the scatter of data is too high.

One important result of this study is the clear separation of tropical trees according to soil type, instead of adapting the usual separation according to life form deciduous and evergreen. The observed difference between tropical trees predominantly grown on 'old' oxisols and those grown on 'young' nonoxisols is plausible, as plants on old oxisol soils usually have a lower ratio of leaf phosphorous content to leaf nitrogen content (P/N ratio), although the variability of leaf P/N ratios is very high between species and individuals on landscape scale (Townsend *et al.*, 2007). Tropical trees on old soils on average are observed to be rather limited by phosphorous and not nitrogen availability (Meir *et al.*, 2007). Thus, leaf nitrogen content should only have a minor impact on photosynthetic capacity (Wardle *et al.*, 2004). This is reflected by the slope of the linear relationship of V_{max}^{25} to leaf nitrogen content, and indeed 'tropical trees on oxisols' is the PFT with the lowest slope. A separation of NUE according to soil age, used as proxy for leaf phosphorous content, which is rarely measured along with photosynthesis rates, may be expected for other PFTs and extra tropical areas as well, but for paucity of data could not be established in this study.

Functional determinants of NUE

In our results, we find a difference by a factor of 5 between the lowest and highest mean NUE per PFT, between tropical trees on oxisols and crops. Compared with these differences of mean NUE, the relative differences between mean values of nitrogen content of different PFTs are small. The variation of mean V_{max}^{25} by PFT is therefore dominated by variations in NUE, while within each PFT, the scatter of V_{max}^{25} values is dominated by the high variability of leaf nitrogen content. Using NUE and V_{max}^{25} as a guide, we can separate the 10 PFTs into three distinct groups (see Fig. 4): one major group of broadleaved trees and shrubs with medium NUE and V_{max}^{25} (PFTs 3, 4, 7, 8); the herbaceous plants and crops with high NUE and V_{max}^{25} (PFTs 9, 10); and a rather heterogeneous group with lower NUE and V_{max}^{25} consisting of conifers and tropical trees (PFTs 1, 2, 5, 6).

The first group has very similar values for both NUE and V_{max}^{25}, hardly affected by either growth form or deciduousness. Within the second, herbaceous group, the higher photosynthetic capacity of crops comes with, in relative terms, and even higher NUE, which shows that the extra performance is not achieved through additional nitrogen, but by more efficient use of resources for photosynthesis. This broad tendency can also be seen at the very general level of PFT groupings. If we extend the linear relationship between NUE and V_{max}^{25} established by PFTs 9 and 10 across Fig. 4, we find the first group, deciduous conifers and the oxisol-tropical trees very close to the interpolated line. Tropical trees on nonoxisols have a lower V_{max}^{25} (or higher NUE, respectively) than this relationship suggests, possibly a sign of adaptation to higher temperatures, avoiding excess respiration costs. While tropical trees on oxisols, due to the potential phosphorous limitation, show low NUE and low V_{max}^{25}.

However, if the effect of deciduousness is considered (which is only small in the first group clustered in the middle, as mentioned), then a different axis of variation appears: the evergreen form always has a higher photosynthetic capacity per area, but a lower NUE. Because we assigned a single set of parameters intercept and slope to both forms to constrain the linear relationship of V_{max}^{25} to leaf nitrogen content (see Eqn (1)), the orientation of this axis results from the positive intercept and the higher leaf nitrogen content per area of the evergreen form, which was found for all three groups. However, it is consistent with the fact that evergreen plants, which here appear to have a lower NUE than their deciduous counterparts, also have a lower specific leaf area (Schulze *et al.*, 1994). This leads to more light extinction within the leaf, lowering photosynthetic capacity per amount of photosynthetic elements, for which NUE is a proxy, but increases photosynthesis on an area basis.

Impact of the new parameterisation on modelled GPP

A comparison of the estimated V_{max}^{25} per PFT to values derived by Beerling & Quick (1995), based on $\delta^{13}C$ values of leaves and A_{max} measurements, modified by Knorr & Heimann (2001), shows major differences. The lower V_{max}^{25} of tropical PFTs in our analysis is reasonable from the viewpoint of optimality assumptions. A low photosynthetic capacity is generally related to low respiration rates. This is an advantage in particular at high night temperatures. The high estimate of V_{max}^{25} for evergreen coniferous trees in our analysis is caused by the exceptionally high leaf nitrogen content per projected leaf area. V_{max}^{25} of shrubs in our analysis is in the same range as broadleaved and coniferous trees, while in the analysis of Beerling & Quick (1995), V_{max}^{25} of deciduous shrubs is higher than for crops. V_{max}^{25} of C3 herbaceous plants in our analysis lies between trees and crops, which seems reasonable, while in Beerling & Quick (1995), V_{max}^{25} of herbaceous plants is about half of our value and in the range of trees. In summary, it appears that the new parameter set is in better agreement with our ecological understanding.

There are only few global parameterisations of V_{max}^{25} and the new dataset developed here significantly differs

© 2009 The Authors

13652486, 2009, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2486.2008.01744.x by Luonnonvarakeskus, Wiley Online Library on [17/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

from the one most often used in global-scale studies. A comparison with independent data at the stand scale shows that the implementation of the new parameter set into the earth system model ECHAM5/JSBACH-BETHY substantially reduces the mismatch between observed and modelled GPP for several forest biomes. While differences in scale and spatial coverage between observed and modelled GPP have to be kept in mind, the excellent agreement for forest types with good data coverage is encouraging.

The reduction of V_{max}^{25} for tropical forests is the most prominent feature of the results and seems to be supported by the observations on stand scale. However, it has to be kept in mind that the number of observations of GPP is low (Table 4) and may contain insufficient data from tropical forests on nutrient poor old oxisol soils. This is important, as about half of the global forest area consists of tropical forests. Besides affecting global carbon turnover, the reduction of V_{max}^{25} has a substantial effect on modelled surface temperature, as in the humid tropical areas the modelled evapotranspiration is limited by maximum stomatal conductance, which directly depends on V_{max}^{25}. According to results of the coupled climate–vegetation model ECHAM5/JSBACH-BETHY (not shown), the reduction of V_{max}^{25} resulted in reduced modelled evapotranspiration and increased surface temperature in the Amazon basin of 0.4–0.5 °C on average. This points towards the importance of a realistic parameterisation of photosynthetic capacity for modelling the climate–vegetation feedback.

Conclusions

We have presented a first approach to assimilate data of V_{max} and A_{max} into the photosynthesis model of Farquhar *et al*. (1980) considering leaf nitrogen content, quantitative plant traits, climate and soil data. The scheme delivers estimates of photosynthetic capacity and its relationship to leaf nitrogen content for different PFTs on a global scale. The analysis, based on a large amount of data, allows the separation of various groups of PFTs according to their photosynthetic capacity and their patterns of nitrogen use. Enough data were included to allow the distinction of tropical trees according to predominant soil type, rather than the common approach of separating into evergreen and deciduous. The reported ranges of V_{max}^{25} were implemented into a global-scale TBM. The newly derived parameterisation could be shown to substantially improve simulated carbon fluxes.

An obvious extension of the approach presented would be the inclusion of a full nitrogen cycle. Leaf nitrogen content as predictor of V_{max}^{25} would not be assigned for each PFTs, but may then respond to changes in N availability, for example, due to changes in mineralisation rates, or to changes in relative allocation patterns between C and N as a consequence of elevated atmospheric CO_2. This is important when considering feedbacks between plant physiology and climate, because, as demonstrated, changes in V_{max}^{25} can have a considerable impact on surface temperature.

Another advantage of the new parameterisation is the inclusion of uncertainty ranges. Parameters presented here can immediately be used as priors in data assimilation schemes of the carbon cycle that employ parameter estimation techniques (Knorr & Kattge, 2005; Rayner *et al*., 2005). The use of prior parameter values based on a large amount of data can be expected to substantially reduce the currently large uncertainty in predictions of the future carbon balance (Friedlingstein *et al*., 2006; Scholze *et al*., 2007).

Acknowledgements

The authors wish to thank Thomas Kaminski for advice and comments, Peter B. Reich for immediately sending additional information about his datasets and Kiona Ogle for help with statistics. This work has been financed and supported by the Max-Planck-Gesellschaft zur Förderung der Wissenschaften, e.V., and the Natural Environment Research Council through its programme Quantifying and Understand the Earth System (QUEST).

References

Bassow SL, Bazzaz FA (1997) Intra- and inter-specific variation in canopy photosynthesis in a mixed deciduous forest. *Oecologia*, **109**, 507–515.

Beerling DJ, Quick WP (1995) A new technique for estimating rates of carboxylation and electron-transport in leaves of C-3 plants for use in dynamic global vegetation models. *Global Change Biology*, **1**, 289–294.

Bernacchi CJ, Singsaas EL, Pimentel C, Portis AR, Long SP (2001) Improved temperature response functions for models of Rubisco-limited photosynthesis. *Plant Cell and Environment*, **24**, 253–259.

Carswell FE, Meir P, Wandelli EV *et al*. (2000) Photosynthetic capacity in a central Amazonian rain forest. *Tree Physiology*, **20**, 179–186.

Coste S, Roggy JC, Imbert P, Born C, Bonal D, Dreyer E (2005) Leaf photosynthetic traits of 14 tropical rain forest species in relation to leaf nitrogen concentration and shade tolerance. *Tree Physiology*, **25**, 1127–1137.

Cox PM, Betts RA, Collins M, Harris PP, Huntingford C, Jones CD (2004) Amazonian forest dieback under climate-carbon cycle projections for the 21st century. *Theoretical and Applied Climatology*, **78**, 137–156.

Domingues TF, Berry JA, Martinelli LA, Ometto JP, Ehlehringer JR (2005) Parameterization of canopy structure and leaf-level gas exchange for an eastern Amazonian tropical rain forest (Tapajós National Forest, Pará, Brazil). *Earth Interactions*, **9**, 1–23.

© 2009 The Authors

13652486, 2009, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2486.2008.01744.x by Luonnonvarakeskus, Wiley Online Library on [17/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Ellsworth DS, Reich PB (1993) Canopy structure and vertical patterns of photosynthesis and related leaf traits in a deciduous forest. *Oecologia*, **96**, 169–178.

Ellsworth DS, Reich PB, Naumburg ES, Koch GW, Kubiske ME, Smith SD (2004) Photosynthesis, carboxylation and leaf nitrogen responses of 16 species to elevated pCO_2 across four free-air CO_2 enrichment experiments in forest, grassland and desert. *Global Change Biology*, **10**, 2121–2138.

Escudero A, Mediavilla S (2003) Decline in photosynthetic nitrogen use efficiency with leaf age and nitrogen resorption as determinants of leaf life span. *Ecology*, **91**, 880–889.

Ethier GJ, Livingston NJ, Harrison DL, Black TA, Moran JA (2006) Low stomatal and internal conductance to CO_2 versus Rubisco deactivation as determinants of the photosynthetic decline of ageing evergreen leaves. *Plant, Cell and Environment*, **29**, 2166–2184.

Evans JR (1989) Photosynthesis and nitrogen relationships in leaves of C_3 plants. *Oecologia*, **78**, 9–19.

Farquhar GD, von Caemmerer S, Berry JA (1980) A biochemical model of photosynthetic CO2 assimilation in leaves of C3 species. *Planta*, **149**, 78–90.

Field C, Mooney HA (1986) The photosynthesis-nitrogen relationship in wild plants. In: *On the Economy of Plant Form and Function* (ed. Givnish T.J.), pp. 25–55. Cambridge University Press, Cambridge, UK.

Franco AC, Bustamante M, Caldas LS *et al*. (2005) Leaf functional traits of Neotropical savanna trees in relation to seasonal water deficit. *Trees*, **19**, 326–335.

Friedlingstein P, Cox P, Betts R *et al*. (2006) Climate – carbon cycle feedback analysis, results from the C4MIP model intercomparison. *Journal of Climate*, **19**, 3337–3353.

Garnier E, Salager JL, Laurent G, Sonie L (1999) Relationships between photosynthesis, nitrogen and leaf structure in 14 grass species and their dependence on the basis of expression. *New Phytologist*, **143**, 119–129.

Grassi G, Meir P, Cromer R, Tompkins D, Jarvis PG (2002) Photosynthetic parameters in seedlings of *Eucalyptus grandis* as affected by rate of nitrogen supply. *Plant Cell and Environment*, **25**, 1677–1688.

Grueters U (1998) *Der Kohlenstoffhaushalt von Weizen in der Interaktion erhöhter CO_2/O_3 Konzentrationen und Stickstoffversorgung*. PhD thesis, Justus-Liebig-University, Gießen, 276 pp. http://geb.uni-giessen.de/geb/volltexte/1999/55/pdf/GruetersUwe-1998-12-18.pdf

Han QM, Kawasaki T, Nakano T, Chiba Y (2004) Spatial and seasonal variability of temperature responses of biochemical photosynthesis parameters and leaf nitrogen content within a *Pinus densiflora* crown. *Tree Physiology*, **24**, 737–744.

Herppich M, Herppich WB, von Willert DJ (2002) Leaf nitrogen content and photosynthetic activity in relation to soil nutrient availability in coastal and mountain fynbos plants (South Africa). *Basic and Applied Ecology*, **3**, 329–337.

Hikosaka K, Nagamatsu D, Ishii HS, Hirose T (2002) Photosynthesis-nitrogen relationships in species at different altitudes on Mount Kinabalu, Malaysia. *Ecological Research*, **17**, 305–313.

Hungate BA, Dukes JS, Shaw MR, Luo Y, Field C (2003) Nitrogen and climate change. *Science*, **302**, 1512–1513.

Kattge J (2002) *Zur Bedeutung von Stickstoff für den CO_2-Düngeeffekt*. PhD thesis, Justus-Liebig-Universität, Gießen, 285 pp. http://geb.uni-giessen.de/geb/volltexte/2002/794/.

Kattge J, Knorr W (2007) The temperature dependence of photosynthetic capacity in a photosynthesis model acclimates to plant growth temperature: a re-analysis of data from 36 species. *Plant Cell and Environment*, **30**, 1176–1190.

Kenzo T, Ichie T, Yoneda R, Kitahashi Y, Watanabe Y, Ninomiya I, Koike T (2004) Interspecific variation of photosynthesis and leaf characteristics in canopy trees of five species of *Dipterocarpaceae* in a tropical rain forest. *Tree Physiology*, **24**, 1187–1192.

Knorr W (2000) Annual and interannual CO_2 exchanges of the terrestrial biosphere: process-based simulations and uncertainties. *Global Ecology and Biogeography*, **9**, 225–252.

Knorr W, Heimann M (2001) Uncertainties in global terrestrial biosphere modeling 1. A comprehensive sensitivity analysis with a new photosynthesis and energy balance scheme. *Global Biogeochemical Cycles*, **15**, 207–225.

Knorr W, Kattge J (2005) Inversion of terrestrial ecosystem model parameter values against eddy covariance measurements by Monte Carlo sampling. *Global Change Biology*, **11**, 1333–1351.

Körner C (1989) The nutritional-status of plants from high-altitudes – a worldwide comparison. *Oecologia*, **81**, 379–391.

Kumagai T, Ichie T, Yoshimura M *et al*. (2006) Modeling CO_2 exchange over a Bornean tropical rain forest using measured vertical and horizontal variations in leaf-level physiological parameters and leaf area densities. *Journal of Geophysical Research-Atmospheres*, **111**.

Larcher W (1994) *Ökophysiologie der Pflanzen*. Verlag Eugen Ulmer, Stuttgart.

Le Roux X, Grand S, Dreyer E, Daudet FA (1999) Parameterization and testing of a biochemically based photosynthesis model for walnut (*Juglans regia*) trees and seedlings. *Tree Physiology*, **19**, 481–492.

Leuning R (1997) Scaling to a common temperature improves the correlation between photosynthesis parameters Jmax and Vcmax. *Journal of Experimental Botany*, **307**, 345–347.

Lusk CH, Reich PB (2000) Relationships of leaf dark respiration with light environment and tissue nitrogen content in juveniles of 11 cold-temperate tree species. *Oecologia*, **123**, 318–329.

Luyssaert S, Inglima I, Jung M *et al*. (2007) CO_2 balance of boreal, temperate, and tropical forests derived from a global database. *Global Change Biology*, **13**, 2509–2537.

Manter DK, Kerrigan J (2004) A/C-i curve analysis across a range of woody plant species: influence of regression analysis parameters and mesophyll conductance. *Journal of Experimental Botany*, **55**, 2581–2588.

Medlyn BE, Badeck FW, De Pury DGG *et al*. (1999) Effects of elevated CO2 on photosynthesis in European forest species: a meta-analysis of model parameters. *Plant, Cell and Environment*, **22**, 1475–1495.

Medlyn BE, Dreyer E, Ellsworth D *et al*. (2002) Temperature response of parameters of a biochemically based model of photosynthesis. II. A review of experimental data. *Plant, Cell and Environment*, **25**, 1167–1179.

Medlyn BE, Jarvis PG (1999) Design and use of a database of model parameters from elevated [CO_2] experiments. *Ecological Modelling*, **124**, 69–83.

© 2009 The Authors

13652486, 2009, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2486.2008.01744.x by Luonnonvarakeskus, Wiley Online Library on [17/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Meir P, Kruijt B, Broadmeadow M *et al*. (2002) Acclimation of photosynthetic capacity to irradiance in tree canopies in relation to leaf nitrogen concentration and leaf mass per unit area. *Plant Cell and Environment*, **25**, 343–357.

Meir P, Levy PE, Grace J, Jarvis PG (2007) Photosynthetic parameters from two contrasting woody vegetation types in West Africa. *Plant Ecology*, **192**, 277–287; doi: 210.1007/s11258-11007-19320-y.

Misson L, Tu KP, Boniello RA, Goldstein AH (2006) Seasonality of photosynthetic parameters in a multi-specific and vertically complex forest ecosystem in the Sierra Nevada of California. *Tree Physiology*, **26**, 729–741.

Morison JL (1987) Intercellular CO_2 concentration and stomatal response to CO_2 (eds Givnish E, Farguher GD, Cowan IR), pp. 229–252. Stanford University Press, Stanford, CA, USA.

Peterson A, Ball J, Luo Y *et al*. (1999) The photosynthesis leaf nitrogen relationship at ambient and elevated atmospheric carbon dioxide: a meta-analysis. *Global Change Biology*, **5**, 331–346.

Quilici A, Medina E (1998) Photosynthesis-nitrogen relationships in pioneer plants of disturbed tropical montane forest sites. *Photosynthetica*, **35**, 525–534.

Raddatz TJ, Reick C, Knorr W *et al*. (2007) Will the tropical land biosphere dominate the climate–carbon cycle feedback during the twenty-first century? *Climate Dynamics*, **29**, 565–574.

Raupach MR, Rayner PJ, Barrett DJ *et al*. (2005) Model-data synthesis in terrestrial carbon observation: methods, data requirements and data uncertainty specifications. *Global Change Biology*, **11**, 378–397.

Rayner PJ, Scholze M, Knorr W, Kaminski T, Giering R, Widmann H (2005) Two decades of terrestrial carbon fluxes from a carbon cycle data assimilation system (CCDAS). *Global Biochemical Cycles*, **19**, GB2026, doi: 2010.1029/2004GB002254.

Reich PB, Ellsworth DS, Walters MB (1998) Leaf structure (specific leaf area) modulates photosynthesis-nitrogen relations: evidence from within and across species and functional groups. *Functional Ecology*, **12**, 948–958.

Reich PB, Ellsworth DS, Walters MB, Vose JM, Gresham C, Volin JC, Bowman WD (1999) Generality of leaf trait relationships: a test across six biomes. *Ecology*, **80**, 1955–1969.

Reich PB, Hungate BA, Luo YQ (2006) Carbon–nitrogen interactions in terrestrial ecosystems in response to rising atmospheric carbon dioxide. *Annual Review of Ecology Evolution and Systematics*, **37**, 611–636.

Reich PB, Kloeppel BD, Ellsworth DS, Walters MB (1995) Different photosynthesis-nitrogen relations in deciduous hardwood and evergreen coniferous tree species. *Oecologia*, **104**, 24–30.

Reich PB, Wright IJ, Lusk CH (2007) Predicting leaf physiology from simple plant and climate attributes: a global GlopNet analysis. *Ecological Applications*, **17**, 1982–1988.

Ripullone F, Grassi G, Lauteri M, Borghetti M (2003) Photosynthesis-nitrogen relationships: interpretation of different patterns between *Pseudotsuga menziesii* and *Populus x euroamericana* in a mini-stand experiment. *Tree Physiology*, **23**, 137–144.

Sage RF, Pearcy RW, Seemann JR (1987) The nitrogen use efficiency of C-3 and C-4 plants. Leaf nitrogen effects on the activity of carboxylating enzymes in *Chenopodium album* (L) and *Amaranthus retroflexus* (L). *Plant Physiology*, **85**, 355–359.

Scholze M, Kaminski T, Rayner PJ, Knorr W, Giering R (2007) Propagating uncertainty through prognostic CCDAS simulations. *Journal of Geophysical Research*, doi: 10.1029/2007JD008642.

Schulze E-D, Kelliher FM, Körner C, Lloyd J, Leuning R (1994) Relationships among maximum stomatal conductance, ecosystem surface conductance, carbon assimilation rate, and plant nitrogen nutrition: a global ecology scaling exercise. *Annual Review of Ecology and Systematics*, **25**, 629–660.

Schulze ED, Lloyd J, Kelliher FM *et al*. (1999) Productivity of forests in the Eurosiberian boreal region and their potential to act as a carbon sink – a synthesis. *Global Change Biology*, **5**, 703–722.

Sitch S, Smith B, Prentice IC *et al*. (2003) Evaluation of ecosystem dynamics, plant geography and terrestrial carbon cycling in the LPJ dynamic global vegetation model. *Global Change Biology*, **9**, 161–185.

Springer CJ, DeLucia EH, Thomas RB (2005) Relationships between net photosynthesis and foliar nitrogen concentrations in a loblolly pine forest ecosystem grown in elevated atmospheric carbon dioxide. *Tree Physiology*, **25**, 385–394.

Tan WX, Hogan GD (1995) Limitations to net photosynthesis as affected by nitrogen status in jack pine (*Pinus banksiana* Lamb) seedlings. *Journal of Experimental Botany*, **46**, 407–413.

Thornton PE, Lamarque J-F, Rosenbloom NA, Mahowald NM (2007) Influence of carbon–nitrogen cycle coupling on land model response to CO_2 fertilization and climate variability. *Global Biogeochemical Cycles*, **21**, GB4018, doi: 10.1029/2006GB002868.

Tissue DT, Griffin KL, Ball JT (1999) Photosynthetic adjustment in field-grown ponderosa pine trees after six years of exposure to elevated CO_2. *Tree Physiology*, **19**, 221–228.

Tissue DT, Griffin KL, Thomas RB, Strain BR (1995) Effects of low and elevated CO_2 on C3 and C3 annuals .2. Photosynthesis and leaf biochemistry. *Oecologia*, **101**, 21–28.

Townsend AR, Cleveland CC, Asner GP, Bustamante MMC (2007) Controls over foliar N : P ratios in tropical rain forests. *Ecology*, **88**, 107–118.

Trudinger C, Raupach M, Rayner P *et al*. (2007) The OptIC project: an intercomparison of optimisation techniques for parameter estimation in terrestrial biogeochemical models. *Journal of Geophysical Research – Biogeosciences*, **112**, G02027, doi: 10.1029/2006JG000367.

Walcroft A, Le Roux X, Diaz-Espejo A, Dones N, Sinoquet H (2002) Effects of crown development on leaf irradiance, leaf morphology and photosynthetic capacity in a peach tree. *Tree Physiology*, **22**, 929–938.

Walcroft AS, Whitehead D, Silvester WB, Kelliher FM (1997) The response of photosynthetic model parameters to temperature and nitrogen concentration in *Pinus radiata* D. Don. *Plant Cell and Environment*, **20**, 1338–1348.

Wardle DA, Walker LR, Bardgett RD (2004) Ecosystem properties and forest decline in contrasting long-term chronosequences. *Science*, **305**, 509–513.

White MA, Thornton PE, Running SW, Nemani RR (2000) Parameterization and sensitivity analysis of the BIOME-BGC

© 2009 The Authors

13652486, 2009, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2486.2008.01744.x by Luonnonvarakeskus, Wiley Online Library on [17/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

terrestrial ecosystem model: net primary production controls. *Earth Interactions*, **4**, 1–85.

Whitehead D, Walcroft AS, Scott NA, Townsend JA, Trotter CM, Rogers GND (2004) Characteristics of photosynthesis and stomatal conductance in the shrubland species manuka (*Leptospermum scoparium*) and kanuka (*Kunzea ericoides*) for the estimation of annual canopy carbon uptake. *Tree Physiology*, **24**, 795–804.

Wohlfahrt G, Bahn M, Haubner E *et al*. (1999) Inter-specific variation of the biochemical limitation to photosynthesis and related leaf traits of 30 species from mountain grassland ecosystems under different land use. *Plant Cell and Environment*, **22**, 1281–1296.

Woodward FI, Smith TM, Emanuel WR (1995) A global land primary productivity and phytogeography model. *Global Biogeochemical Cycles*, **9**, 471–490.

Wright IJ, Reich PB, Westoby M *et al*. (2004) The worldwide leaf economics spectrum. *Nature*, **428**, 821–827.

Wullschleger SD (1993) Biochemical limitations to carbon assimilation in C3 plants – a retrospective analysis of the A/Ci curves from 109 species. *Journal of Experimental Botany*, **44**, 907–920.

Yoshie F (1986) Intercellular CO_2 concentration and water-use efficiency of temperate plants with different life-forms and from different microhabitats. *Oecologia*, **68**, 370–374.

Zaehle S, Sitch S, Smith B, Hatterman F (2005) Effects of parameter uncertainties on the modeling of terrestrial biosphere dynamics. *Global Biogeochemical Cycles*, **19**, GB3020, doi: 10.1029/2004GB002395.

© 2009 The Authors

13652486, 2009, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1365-2486.2008.01744.x by Luonnonvarakeskus, Wiley Online Library on [17/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License